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Why the need for Confidential AI is on the rise: Intel's Anand Pashupathy explains

It's the principles of Confidential AI, that will govern how this transformative technology progresses and evolves securely, ethically, and is beneficial for all

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If 2023 was all about getting enamoured and captivated by the rise of Generative AI left, right and centre, then 2024 and beyond is most definitely about securing AI—from chip to software applications and beyond—to the maximum. In fact, in today's AI dominated tech landscape, maintaining the confidentiality and privacy of data has become a paramount concern, since digital threats loom as prominently as innovations. Amidst this backdrop, I unpacked the concept of Confidential AI with Anand Pashupathy, a key figure at Intel driving the frontier of

secure AI workloads and applications. Our discussion highlighted an emerging paradigm that promises to reshape how we manage and secure data in the age of ubiquitous AI.

What is Confidential AI

Anand Pashupathy, Vice President & General Manager, Security Software & Services Division at Intel, begins by explaining the inception and necessity of Confidential AI, a term that may still sound like jargon even to the seasoned ears. "Intel looks at AI and security as kind of being at this intersection," Anand

remarks. The intersection he refers to is not just about using AI for enhancing security measures or securing AI itself—it's about a holistic approach to safeguard data during AI operations.

Confidential AI emerges from the broader concept of confidential computing, which focuses on protecting data while it's being processed, not just when it's stored or in transit. In simpler terms, it's about ensuring sensitive data privacy throughout the AI lifecycle, from training to inference. Anand illustrates, "Confidential computing ensures that data is opaque inside a trusted execution environment, making it secure from external threats even during computation." This foundation is pivotal for AI systems where data often traverses and transforms through various states and operations.

Delving deeper, Anand explains the mechanics of how Confidential AI works, highlighting its role in securing AI operations that span across industries, from healthcare to financial services. He talks about a scenario that is becoming increasingly common: AI models being tampered with, which could lead to disastrous outcomes. "About 86 percent of enterprises are concerned about the security of their models," he suggests, based on a Forrester report, underscoring the widespread data-security related anxiety that plagues industries relying on AI.

Confidential AI addresses these fears head-on by implementing layers of security within the computing processes that host and handle AI models. This involves sophisticated encryption techniques not just when the data is at rest or in motion, but crucially, when it is active and being processed. "When data is operated upon, it has to be unencrypted traditionally, which is where the vulnerability creeps in. Confidential AI counters this by maintaining encryption even during computation within a secure environment," Anand explains.

Confidential AI: Real-world applications and implications

The conversation shifts to the real-world applications of Confidential AI, where